

Solution Requirements

Project Title: Rising Waters

Introduction

The **Rising Waters** system requires both hardware and software components to successfully predict flood risk using Machine Learning. These requirements ensure smooth development and execution of the application.

Functional Requirements

- Accept weather parameters from the user.
- Process the input data.
- Predict flood or no flood using the trained ML model.
- Display the prediction along with probability.
- Provide a simple and user-friendly interface.

Non-Functional Requirements

- Fast prediction response.
- High prediction accuracy.
- Easy to use and maintain.
- Reliable and secure application.

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Storage: 10 GB free space
- Internet connection (optional)

Software Requirements

- Operating System: Windows 10/11
- Python 3.x
- Flask
- Pandas
- NumPy

- Scikit-learn
- XGBoost
- Joblib
- HTML, CSS, JavaScript
- Visual Studio Code or PyCharm

Conclusion

The above requirements provide the necessary environment for developing and running the **Rising Waters** Flood Prediction System efficiently.